Table showing cutting speed for various materials

Work Material	Cutting tool material		
	High speed steel	Tungsten steel	Stellite
Mild steel	30 m/min	80 m/min	58 m/min
High carbon steel	26 m/min	65 m/min	50 m/min
Cast steel	15 m/min	80 m/min	42 m/min
Cast iron	22 m/min	80 m/min	50 m/min
Aluminium	90 m/min	400 m/min	330 m/min
Brass	61 m/min	200 m/min	33 m/min

Tools used in a lathe

A. According to the construction. The lathe tools are classified into three types

1. Solid tool
2. Brazed tipped tool
3. Tool bit and tool holders

B. According to the operation to be performed, the cutting tools are classified as

1. Turning tool
2. Thread cutting tool
3. Facing tool
4. Forming tool
5. Parting tool
6. Grooving tool
7. Boring tool
8. Internel thread cutting tool
9. Knurling tool

C. According to the direction of feed movement, the following tools are used

1. Right hand tool
2. Left hand tool
3. Round nose tool

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